



Family Medical Officer Test Plan

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1 INTRODUCTION

1.1 Purpose

This is the comprehensive test plan of the Family Medical Officer project. The purpose of the document describes scopes of test and activities which need to be taken during test process of project. It addresses the following items:

- Scopes of Testing
- Requirements for Testing
- Test Strategy
- Test Resources
- Test Milestones
- Test Deliverables

1.2 Definitions, Acronyms, and Abbreviations

Abbreviations	Description
AT	Acceptance Test
DMS	Defect Management System (a FPT in-house tool)
IT	Integration Test
ST	System Test
PM	Project Manager
PTL	Project Technical Leader
QA	Quality Assurance
FRS	Functional Requirement Specification
TP	Test Plan
TC	Test Case
TR	Test Report
UAT	User Acceptance test
UT	Unit Test
ECH	El Camino Hospital
HIPAA	Health Insurance Portability and Accountability Act
TRUSTe	Privacy Seals, Online Trust & Safety service
WCG World	The third-party who will be with ECH to do UI Design

1.3 References

Title/File name	Author	Version	Effective Date
ECH_MobileWeb_0722.pdf	WCG	4	07/26/11
iPhone_App_Concepts_Presentation_by_ECH_201101.pdf	WCG		01/01/11
iPhone_App_Research_Presentation_by_ECH_20110126.pdf	WCG		01/26/11
Mobile_Optimized_Website_Strategy_Presentation_v1_20110328.pdf	WCG	1	03/28/11
Mobile_POV_v2_20110208.pdf	WCG	2	02/08/11
Project_History_Timeline_20110713.pdf	FPT		07/13/11
FMO_Questions and Answers Management Sheet_20110722.xlsx	FPT		07/26/11
FMO_WO_v1.0.doc	FPT	1.0	07/22/11
FMO_Project Plan_v1.0.doc	FPT	1.0	07/11/11
FMO_Functional Requirement Specification_v1.0	FPT	1.0	07/27/11

1.4 Background information

El Camino Hospital (ECH) is a not-for-profit organization with hospital campuses in Mountain View and Los Gatos, California. The hospitals have served communities in the South San Francisco Bay Area for nearly 50 years.

ECH has a website that is currently serving the desktop users. In order to better serve the mobile consumers with a many different kinds of platforms such as iPhone and Android mobile devices, ECH is in need of new version to serve the mobile users. This new version has the following main functions:

- Emergency Room
- Find a Physician
- My Family & Me
- ECH Resources
- Visiting ECH
- ECH News

The purpose of this project is to develop a HTML5 based web application so that it can function well on both iPhone and Android mobile phones then develop the iPhone and Android applications to link with the HTML5 web app.

1.5 Scope of testing

The scope of test will be limited to testing three applications: the HTML 5 applications and two hybrid applications (on iPhone and Android).

A. Target of Test

Functional items and Non-functional items will be verified and passed by FPT development team, then be validated and approved by El Camino Hospital via test stages, including the requirements of the following primary functions:

- Emergency Room - Find a Physician - My Family & Me	- ECH Resources - Visiting ECH - ECH News
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B. Test Stage

No	Test Stages	Description
1	Unit Test	Unit Test will be performed by FPT development team.
2	Integration Test	Integration Test will be performed by FPT QA team. After the Unit Test is finished, testers will execute the UT Gate based on the UT Gate checklist for each function. Integration Test will only start if the result of UT Gate is Passed. This test stage focuses on specific areas of use cases when all requirements are completed, integration test should be performed to ensure all components incorporate well.
3	System Test	System Test will be executed by FPT QA team. Testers will perform complete, end-to-end system testing staged in pre-production environment to validate that functions and system interfaces perform properly in production environment.
4	Acceptance Test	Acceptance Test will be conducted by El Camino Hospital.

No	Test Stages	Description
		FPT's responsibilities during Acceptance Test phase are: • Facilitate completion on the application deployment • Support fixing bugs • Support Final User Acceptance Test

C. Test Assumption

The following assumptions are made for test process:

- **El Camino Hospital is to validate and approve for final software product, test procedures and results**
- Verification from FPT project team for test execution, documenting and results
- FPT project managers to approve and commit for test resources
- Requirements for test are limited to functional and non-functional requirements specified in Section 2 of this document
- Test will be executed on specific hardwares and softwares as defined in Section 4.2

1.6 Constraints

The following constraints may apply when testing is performed on system:

- Deadline for testing only can be met if development progress is on time
- Test execution can be performed when system passes Unit Test Inspection
- At least one round of testing must be performed for requirements
- Complete testing needs support from El Camino Hospital for production environment set-up

1.7 Risk List

#	Description	Source	Probability	Exposure	Trigger
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#	Description	Source	Probability	Exposure	Trigger
1	Depends on the UI design and requirement documents from WCG World	Critical Dependencies	0.7	5.6	

1.8 Training Needs

No	Training items	Description
1	Unit Test	Learn & study from internal: FPT's other teams Learn & study from external: 1. Microsoft Pax: http://research.microsoft.com/en-us/downloads/d2279651-851f-4d7a-bf05-16fd7eb26559/default.aspx 2. W3school validator: http://validator.w3.org/mobile/ 3. Microsoft Visual Studio 10 ultimate
2	System Test	Learn & study from external: 1. Load test: Open STA, Jmeter, IBM Rational Robot
3	Security Test	Learn & study from external: https://www.owasp.org/ 1. Ethereum/WireShark 2. http://www.wedi.org/snip/public/articles/testing_whitepaper_082602.pdf 3. http://www.macadamian.com/images/uploads/whitepapers/HIPAA_TestStrategies.pdf

2 REQUIREMENTS FOR TEST

2.1 Test Items

A. Functional Items

1. Homepage	12. My Family & Me - View Profile
2. ER Wait Time	13. My Family & Me - Edit Profile
• View wait time	14. My Family & Me - Delete Profile/ Account

• Call Emergency Room • Call Nurses' Station • Get Directions	• Call main line • Get Directions • View Campus Map • Visitor Information • Get Directions nearby
3. ER Checklist	15. My Family & Me- Delete Account
4. Find a Physician By Last Name	16. ECH Resources - Vital Signs
5. Find a Physician By Specialty	17. ECH Resources - Common Warning Signs
6. Find a Physician By Location	18. Visiting ECH - Pt 1 of 2
7. My Family & Me - Login	
8. My Family & Me - Forgot Password	
9. My Family & Me - Dashboard	19. Visiting ECH - Pt 2 of 2
10. My Family & Me - Create An Account	• Get Directions nearby
11. My Family & Me - Create a Profile	20. ECH News

B. Non functional Items

1. System must be HIPAA and TRUSTe compliance.
2. In the normal condition (could be 100 concurrent users or less), each page should load in 4 seconds or less.
3. In the stress condition (could be more than 100 concurrent users), each page should load in 12 seconds or less.

2.2 Acceptance Test Criteria

No	Test Stages	Qualified ratios
1	Unit Test	To pass this stage, all unit test cases must be tested and passed 100%. All defects should be fixed and re-tested. Average of 11 bugs/KLOC.
2	Integration Test	To pass this stage, all test cases must be tested and passed 100%. All defects should be fixed and re-tested. Average of 4 bugs/KLOC.
3	System Test	To pass this stage, all test cases must be tested and passed 100%. All defects should be fixed and re-tested. Average of 0.5 bugs/KLOC.
4	Acceptance Test	Acceptance Test will be conducted and approved by El Camino Hospital.

1	Functional requirement	100% of all requirement defined on part 2.1 A
2	Non-functional requirement	100% of all requirement defined on part 2.1 B

3 TEST STRATEGY

3.1 Test Types

A. Function Testing

Test Objective:	Verify the application and its internal processes by interacting with the application via the Graphical User Interface (GUI) and analyzing the outputs or results
Technique:	- Testers will create test scenarios against the requirements provided by customer. Test scenarios will be created based on black box test technique. - Testers execute test based on test scenarios and create report. Common defects will be collected for improved checklist. - Execute each case, using valid and invalid data, to verify the following: Get the expected results when valid and invalid data is used Valid input data is updated correctly into database The appropriate errors or warning messages are displayed when invalid data is used - Execute each case, using boundary data, to verify the following: Get the expected results when boundary data is used Data is updated correctly to database The appropriate errors or warning messages are displayed when invalid data is used - Each valid data input is updated correctly into database. - Each business rule is properly applied.
Completion Criteria:	All functional test cases have been executed to verify proper data acceptance, processing, and retrieval, and the appropriate implementation of the business rules, and passed The appropriate activities will be performed when valid data is used The corresponding error/warning message mechanism is applied for each specific case

	All bugs found must be fixed
Special Considerations:	Functional testing will NOT be started in case of developers have not executed unit test before passing application to testers

B. User Interface Testing

Test Objective:	Navigation through the target-of-test properly reflects business functions and requirements, including screen to screen, field-to-field, and use of access methods. Objects and characteristics, such as menus, size, position, state, and focus conform to standards.
Technique:	- Testers will create specific test scenarios against UI prototypes approved by customer. - Testers execute test based on test scenarios and create report. Common defects will be collected for improved checklists. - Execute each case, using valid, invalid and boundary data, to verify the expected results display when valid, invalid and boundary data is used.
Completion Criteria:	- All GUI test cases have been executed and passed - Any defects related to GUI are fixed, except those accepted by customer - All defects found by tester are addressed in DMS
Special Considerations:	N/A

C. Load Testing

Test Objective:	Verify the application and its internal processes should be met: In the normal condition could be 100 concurrent users or less, each page should load in four seconds or less In the stress condition could be more than 100 concurrent users, each page should load in 12 seconds or less
Technique:	- Testers will create test scenarios, test scripts against the requirements provided by customer. Test scenarios will be created based on black box test technique, and be supported by one of the following tools: IBM Rational Robot & Manager. - Testers execute test based on test scenarios and create report. Common defects will be collected for improved checklists. - Execute each case, using valid and invalid data, to verify the following on a random device:

	In the normal condition could be 100 concurrent users or less, each page should load in four seconds or less In the stress condition could be more than 100 concurrent users, each page should load in 12 seconds or less
Completion Criteria:	In the normal condition could be 100 concurrent users or less, each page should load in four seconds or less. In the stress condition could be more than 100 concurrent users, each page should load in 12 seconds or less All performance requirements must be met
Special Considerations:	Numbers of 100 virtual users should be available for Microsoft Visual Studio

D. Security Testing

Test Objective:	Verify that the application is HIPAA & TRUSTe compliance
Technique:	- Testers will create test scenarios against the requirements which are based on HIPAA & TRUSTe compliance. Test scenarios will be created based on black box test technique. Refer to: https://www.owasp.org/; http://www.wedi.org/snip/public/articles/testing_whitepaper082602.pdf; http://www.macadamian.com/images/uploads/whitepapers/HIPAA_TestStrategies.pdf for more detail. - Testers execute test based on test scenarios and create report. Use WireShark tool to validate transaction encrypted or not. Common defects will be collected for improved checklists. - Execute each case, using valid and invalid data, to verify the following: The expected results occur when valid HIPAA & TRUSTe compliance.
Completion Criteria:	All test cases have been executed to verify proper data acceptance, processing, and retrieval, and the appropriate implementation of the HIPAA & TRUSTe compliance rules, and passed
Special Considerations:	HIPAA & TRUSTe compliance requirement

E. Regression Testing

Test Objective:	Verify the application on new build/ after bug fixing, and be sure that other functions is not affected by fixed parts each iterations
Technique:	- Testers will select some test scenarios against functional test - Testers execute test based on test scenarios and create report. Common defects

	will be collected for improved checklists. - Execute each case, using valid and invalid data, to verify the following: Get the expected results when valid and invalid data is used Valid input data is updated correctly into database The appropriate errors or warning messages are displayed when invalid data is used - Execute each case, using boundary data, to verify the following: Get the expected results when boundary data is used Data is updated correctly into database The appropriate errors or warning messages are displayed when invalid data is used - Each valid data input is updated correctly into database. - Each business rule is properly applied
Completion Criteria:	Specified function test cases have been executed to verify proper data acceptance, processing, and retrieval, and the appropriate implementation of the business rules, and passed The appropriate activities will be performed when valid data is used. The corresponding error/warning message mechanism is applied for each specific case. All bugs found must be fixed. Some of other functions is not affected by fixed code
Special Considerations:	N/A

3.2 Test Stages

Types of Test	Stages of Test			
	Unit	Integration	System	Acceptance
Function Test	X	X	X	X
User Interface test	X	X	X	X
Load test			X	X
Security test	X		X	X
Regression test	X	X	X	

3.3 Tools

Purpose	Tool	Vendor/In-house	Version
Defect log	DMS	FPT tool	
Collect test effort	Timesheet	FPT tool	
Documenting	MS Word, Excel	Microsoft	2010
Unit Testing	Pax	Microsoft	
Function Testing	Manual	Manual	
Load Testing	IBM Rational Robot & Manager	Microsoft	2010
Security Testing	WireShark	<open source>	1.6.1
Database	SQL Server	Microsoft	2008

4 RESOURCE

4.1 Human Resource

This table shows the staffing assumptions for the project.

Worker/Doer	Role	Specific Responsibilities/Comments
Anh Pham	Test Leader	Manage test resources and assign test tasks Create Test Plan, Test Cases (IT, ST), Test Scripts (IT, ST) Review Test Data Create Test Reports
Hien Nguyen	Tester	Review Test Cases (IT, ST) Create Test Data and Execute Test (IT, ST) Report Test Results
Linh Doan	Quality Assurance	Final Inspection Test Cases, Test Plan, Test Reports
Chien Nguyen	Project Manager	Approve Test Cases (UT, IT, ST), Test Plan, Test Results, Test Reports
Nam Le	Project Technical Leader	Review Test Cases (UT, IT, ST)
Anh Vu	Developer	Create UT Cases, UT Reports
Tung Tran	Developer	Create UT Cases, UT Reports
Dat Nguyen	Developer	Create UT Cases, UT Reports

4.2 Hardwares and Softwares

The FMO HTML5 web-based application and iPhone, Android Hybrid applications will require testing on the following iPhone and Android devices:

Hardware	OS Version
iPhone 4	iOS 4
Motorola DROID 2	Android OS, v2.2 (Froyo)
Samsung Galaxy S II	Android OS, v2.3 (Gingerbread)
HTC Sensation	Android OS, v2.3 (Gingerbread)

Software	Version
Microsoft Windows Server	2003 SP2
Microsoft IIS	6.0
Microsoft SQL Server	2008
Microsoft .NET Framework	4.0

5 TEST MILESTONES

Milestone Task	Effort (pd)	Start Date	End Date
Create Test Plan	5	07/15/11	07/26/11
Review & Update Test Plan	2	07/26/11	07/27/11
Create IT&ST Test Cases	13	07/27/11	08/12/11
Review & Update IT&ST Test Cases	2	08/09/11	08/15/11
Create & Execute & Report UT for Software Package version 1.0	15	08/08/11	09/05/11
Create test data for Software Package version 1.0	1	08/31/11	09/02/11
UT Gate for Software Package version 1.0	0.5	09/05/11	09/06/11
Execute IT for Software Package version 1.0	16	09/05/11	09/17/11
Execute ST for Software Package version 1.0	4	09/17/11	09/19/11
Create IT&ST Test Report for Software Package version 1.0	1	09/20/11	09/20/11
Create & Execute & Report UT for	22	09/19/11	10/10/11

Software Package version 1.1			
UT Gate for Software Package version 1.1	0.5	10/11/11	10/11/11
Create test data for Software Package version 1.1	1	10/05/11	10/08/11
Execute IT for Software Package version 1.1	17	10/10/11	10/22/11
Execute ST for Software Package version 1.1	5	10/22/11	10/24/11
Create IT&ST Test Report for Software Package version 1.1	1	10/24/11	10/24/11
Create & Execute & Report UT for Software Package version 1.2	16	09/10/11	10/04/11
UT Gate for Software Package version 1.2	0.5	10/20/11	10/20/11
Create test data for Software Package version 1.2	1	10/17/11	10/19/11
Execute IT for Software Package version 1.2	20	10/20/11	11/04/11
Execute ST for Software Package version 1.2	3	11/04/11	11/07/11
Create IT&ST Test Report for Software Package version 1.2	0.5	11/07/11	11/07/11
Execute System Test for Software Package final version	8	11/03/11	11/12/11
Create Test Report for Software Package final version	1	11/14/11	11/14/11

6 DELIVERABLES

No	Deliverables	Delivered Date	Delivered by	Delivered to
1	Test Plan	08/01/11	FPT	ECH
2	Test Reports for Software Package version 1.0	9/20/11	FPT	ECH
3	Test Reports for Software Package version 1.1	10/24/11	FPT	ECH
4	Test Reports for Software Package version 1.2	11/7/11	FPT	ECH
5	Test Reports for Software Package final version	11/17/11	FPT	ECH